

Q1) Structure factor and intensity calculations (30 pts)

- Calculate the structure factor of a cubic F crystal with a single atom basis and identify its systematic absences.
- Derive the relationship for the interplanar spacing of the cubic F unit cell, with lattice parameter, a .
- Using Bragg's law and assuming $\lambda = 1.54 \text{ \AA}$ and $a = 5.64 \text{ \AA}$, calculate the ten lowest Miller index interplanar spacings and diffraction angles.
- Calculate the structure factor of NaCl and compute the intensity, I_{hkl} .
- Identify the systematic absences of NaCl. How do they differ from those of the cubic F lattice?
- Using (a) and (d), compare the intensity of the first five diffractions present in the cases of cubic F and NaCl.
- Discuss how the intensities in (f) may be used to distinguish an elemental cubic F from a binary cubic F, such as NaCl.

Q2) Designing an XRD experiment to detect phase transition (10 pts)

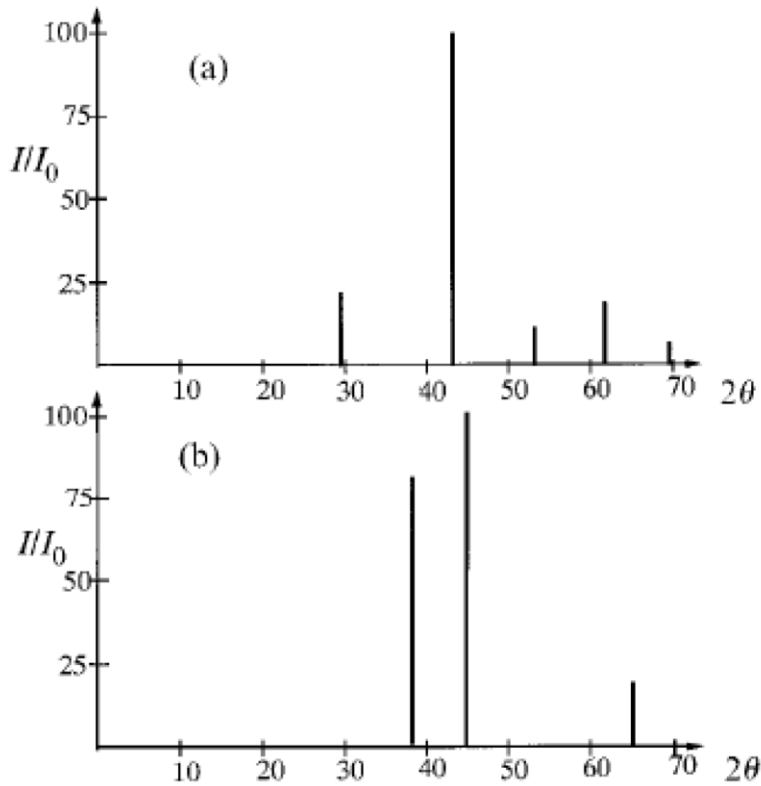
BaTiO_3 has a structure that is completely analogous to SrTiO_3 , except that it is tetragonal. Assume that SrTiO_3 and BaTiO_3 form solid solutions, $(\text{Sr,Ba})\text{TiO}_3$, but at an undetermined concentration of Ba, the solid solution takes on a tetragonal structure. Outline a powder diffraction experiment that could be used to determine the concentration of Ba at which the material transforms from cubic to tetragonal. Be sure to explain the signature that the transformation would have on the diffraction experiment.

Q3) Compound identification (20 pts)

Consider the reaction of Al powder and Fe powder to form FeAl. The structural data for these materials are given in the table below.

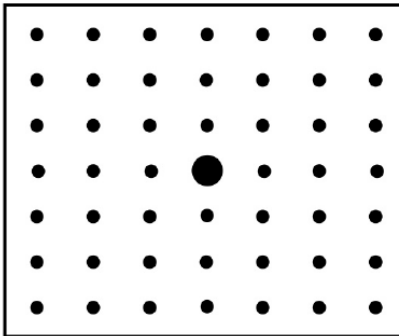
Material	space group	lattice param.	structure type
Fe	$Im\bar{3}m$	$a_0 = 2.87 \text{ \AA}$	bcc
Al	$Fm\bar{3}m$	$a_0 = 4.08 \text{ \AA}$	fcc
FeAl	$Pm\bar{3}m$	$a_0 = 3.01 \text{ \AA}$	B2 or CsCl

One of the schematic diffraction patterns shown below is of the combined, unreacted powders of Al and Fe, and the other is of the intermetallic phase, FeAl. Distinguish between the two. All computations and reasoning on which you base your decision must be included with your answer. Assume that the X-ray wavelength is $1.54 \text{ \AA}$ and the experimental resolution is limited to $\pm 0.5^\circ$ in 2θ .



Q4) Diffraction pattern (30 pts)

Using the reciprocal lattice pattern, draw the $hk0$, $h0l$, and $0kl$ sections of the diffraction patterns of the following space groups considering all systematic absences; explain your answer:



- a) $I4_1/a$
- b) $Cmc2_1$
- c) $Aba2$

Q5) Diffraction Intensity (10 pts)

List all the factors that can influence the intensity of a diffraction intensity.